# Speaker script: Centrality-Based Epidemic Control in a Hospital Ward Contact Network

Group 17 · STAT 636, Fall 2026 · Talk on Tuesday, November 17, 2026, in class

Generated from the speaker notes of the slide sources (deliverables/slides_src). Timing assumes 140 words per minute. Target: 12–13 minutes of talk, then 2–3 minutes of questions. All three members speak.

## Timing summary

| Slide | Title | Speaker | Words | Minutes |
|---|---|---|---|---|
| 1 | Centrality-Based Epidemic Control in a Hospital Ward Contact Network | Arvinder Singh Mundra | 66 | 0.5 |
| 2 | Isolation capacity is scarce. Who should go first? | Arvinder Singh Mundra | 148 | 1.1 |
| 3 | Four days of sensor contacts in a geriatric ward | Arvinder Singh Mundra | 225 | 1.6 |
| 4 | Four centrality measures, four different rankings | Hasini Kavuru | 181 | 1.3 |
| 5 | A weighted SIR model on daily contact hours | Hasini Kavuru | 206 | 1.5 |
| 6 | Every centrality rule beats random isolation | Hasini Kavuru | 142 | 1.0 |
| 7 | The best rule depends on how many can be isolated | Satvika Reddy Akati | 168 | 1.2 |
| 8 | Hiding up to 40% of contacts barely hurts | Satvika Reddy Akati | 177 | 1.3 |
| 9 | Interpreting the pattern: communities and rewiring | Satvika Reddy Akati | 218 | 1.6 |
| 10 | Open questions, and how to address them | Hasini Kavuru | 94 | 0.7 |
| 10 | Open questions, and how to address them | Arvinder Singh Mundra | 144 | 1.0 |
|  | Total |  | 1769 | 12.6 |

| Speaker | Words | Minutes | Share |
|---|---|---|---|
| Arvinder Singh Mundra | 583 | 4.2 | 33% |
| Hasini Kavuru | 623 | 4.5 | 35% |
| Satvika Reddy Akati | 563 | 4.0 | 32% |

## Script by slide

### Slide 1: Centrality-Based Epidemic Control in a Hospital Ward Contact Network

**Speaker:** Arvinder Singh Mundra · **66 words, about 0.5 min** (cue in notes: Arvinder (about 30 seconds))

Hi everyone, we are Group 17: I am Arvinder, and with me are Hasini and Satvika. Our project asks a practical question. When an infection enters a hospital ward and only a few people can be isolated, who should they be? To answer it, we use a real contact network recorded by wearable sensors, and we simulate outbreaks on it to compare network-based ways of choosing.

**Handover / cue line:** (continues to the next slide, same speaker)

## Slide 2: Isolation capacity is scarce. Who should go first?

**Speaker:** Arvinder Singh Mundra · **148 words, about 1.1 min** (cue in notes: Arvinder (about 65 seconds))

In a hospital, isolation beds and staff time are limited, so the question is who to isolate first. The obvious answer is the busiest people, those with many contacts or long contact time. Degree and eigenvector centrality capture that idea. But a person can also matter because they are a bridge: they sit on the short routes between people or groups that rarely meet, even if they are not the busiest. Betweenness and harmonic centrality capture that. These measures often rank people differently, so it is not obvious which one a hospital should use. Our primary question is: which centrality-based isolation rule most reduces the expected outbreak size, and does it still work when contact information is incomplete, as it always is in a real outbreak? One caveat up front: we model preventive isolation before the infection arrives, so this is an idealized intervention, not a clinical policy.

**Handover / cue line:** (continues to the next slide, same speaker)

## Slide 3: Four days of sensor contacts in a geriatric ward

**Speaker:** Arvinder Singh Mundra · **225 words, about 1.6 min** (cue in notes: Arvinder (about 95 seconds), then hand over to Hasini)

Our data come from the SocioPatterns project. Seventy-five people in a geriatric ward in Lyon wore proximity sensors for four days in December 2010: 29 patients, 27 nurses and aides, 11 doctors and 8 administrators. Each record is a 20-second interval in which two people were within about one and a half metres, and there are 32,424 of them. We downloaded the file directly from sociopatterns.org and checked it against these published counts. We aggregate the records into a weighted network: two people are linked if they ever met, and the weight is their average daily contact hours. The network is very dense. It has 1,139 edges, a density of 0.41 and a diameter of just 3, so almost everyone is within two steps of everyone else. Contact time is very uneven: the median person has about 0.6 hours a day and the busiest nurse almost 6. In the picture, colour is role and node size is daily contact hours. Who mixes with whom also matters for spread. Patients spend 75 percent of their contact time with nurses and only 5 percent with other patients, while doctors spend 75 percent of theirs with other doctors. So nurses are the main route between patients, and doctors form their own cluster. That is why the choice of measure could matter here. Hasini will now explain our methods.

**Handover / cue line:** Hasini will now explain our methods.

## Slide 4: Four centrality measures, four different rankings

**Speaker:** Hasini Kavuru · **181 words, about 1.3 min** (cue in notes: Hasini (about 80 seconds))

Thanks, Arvinder. We compare four standard centrality measures, computed in igraph. Degree counts distinct contacts; it is the simplest rule a ward could use. Eigenvector centrality uses contact hours and rewards being in contact with people who are themselves well connected. Betweenness and harmonic centrality use shortest paths, where we set the length of an edge to one over its contact hours, so long contacts are short distances. We use harmonic instead of closeness because it is still defined when the network falls apart, which matters once we hide edges later. The measures disagree. Across all pairs, the rank correlations go from 0.59 to 0.95. On the right, eigenvector and harmonic lie almost on a line, while degree and betweenness are scattered. Degree's top pick is an administrator with 61 distinct contacts but only two hours of contact a day, while the weighted measures all start with nurses. Each rule isolates its top k people, ranked once, with ties broken by a fixed random order. Our baseline is k people chosen at random, with a new set for every simulated outbreak.

**Handover / cue line:** Thanks, Arvinder.

## Slide 5: A weighted SIR model on daily contact hours

**Speaker:** Hasini Kavuru · **206 words, about 1.5 min** (cue in notes: Hasini (about 90 seconds))

To simulate outbreaks we wrote our own discrete-time SIR model in R. One step is one day. Each infected person independently infects a susceptible contact with probability one minus exp of minus lambda times their daily contact hours, and that combines into the formula shown. People infected at the start of a day transmit and then recover with probability 0.2, so they are infectious for five days on average, and newly infected people start transmitting the next day. Before using it, we checked the simulator against exact answers in eight cases, including the full outbreak-size distribution on a small network, computed by enumerating every possible path.

Lambda, the transmission intensity per contact hour, is not known for a specific pathogen, so we chose three values from pilot runs without any isolation: about 25, 50 and 75 percent of the ward infected. We fixed them before comparing strategies. We then isolate the top 1, 3, 5 or 10 people and start each outbreak from one random person who is not isolated. The outcome is the attack rate, the share of all 75 people ever infected. We planned 500 runs per condition but used 3,000, because with 500 the intervals were too wide to tell the measures apart.

**Handover / cue line:** (continues to the next slide, same speaker)

### Slide 6: Every centrality rule beats random isolation

**Speaker:** Hasini Kavuru · **142 words, about 1.0 min** (cue in notes: Hasini (about 60 seconds), then hand over to Satvika)

This is our main result. Each panel is one transmission level. The x-axis is how many people we isolate, and the y-axis is the mean attack rate with 95 percent intervals. The grey line is random isolation, and the four coloured lines are the centrality rules. For three or more people, every centrality rule is clearly below random, at all three levels. With 10 people isolated, the low scenario drops from 17.5 percent infected under random isolation to between 5 and 7 percent. The medium scenario drops from 39 percent to between 19 and 24 percent. Even in the high scenario it drops from 61 to around 50 percent. Isolating a single person barely matters, at most about 3 points. Notice that the coloured lines are close together, so which centrality is best needs a closer look. Satvika will take that next.

**Handover / cue line:** Satvika will take that next.

### Slide 7: The best rule depends on how many can be isolated

**Speaker:** Satvika Reddy Akati · **168 words, about 1.2 min** (cue in notes: Satvika (about 70 seconds))

Thanks, Hasini. Because the centrality curves were close, we zoomed in. This plot shows each weighted measure's attack rate minus that of degree-based isolation. It is pooled over five independent sets of 3,000 runs, so 15,000 per condition, and below zero means better than degree. Two patterns held in at least four of the five replicates. First, when only three to five people can be isolated, at low and medium transmission, betweenness, eigenvector and harmonic all beat degree by one to three percentage points, and every interval excludes zero. The reason is visible in the rankings: degree spends one of its few picks on the administrator who meets many people only briefly. Second, at ten people the pattern flips. Degree has the lowest attack rate at every level, and harmonic is two to five points worse. At high transmission and small k, the rules are within about one point of each other. Differences that small sometimes changed sign between replicates, so we only report patterns that held consistently.

**Handover / cue line:** Thanks, Hasini.

### Slide 8: Hiding up to 40% of contacts barely hurts

**Speaker:** Satvika Reddy Akati · **177 words, about 1.3 min** (cue in notes: Satvika (about 75 seconds))

Real contact tracing is always incomplete. So we hid 20 or 40 percent of the edges at random, computed the centralities and chose whom to isolate on what remained. We then simulated spread on the true, full network, because the infection follows the real contacts. We used 30 missing-edge samples per level, shared by all the rules, with 100 outbreaks each. Our intervals use the spread across those 30 samples, so they include both the randomness of the epidemic and the randomness of which contacts are missing. The plot shows ten people isolated. The surprise is the contrast between the two numbers on the right. Who gets picked changes a lot: only 30 to 92 percent overlap with the complete-data choice. But how well it works hardly changes. Attack rates move by at most about three points in either direction, and every centrality rule stays clearly ahead of random isolation. Our interpretation is that this ward has many similar, heavily connected nurses, so losing some contacts swaps one busy nurse for another with nearly the same effect.

**Handover / cue line:** (continues to the next slide, same speaker)

### Slide 9: Interpreting the pattern: communities and rewiring

**Speaker:** Satvika Reddy Akati · **218 words, about 1.6 min** (cue in notes: Satvika (about 95 seconds), then hand over to Hasini)

Why do the rules behave this way? We looked at two things. First, normalized Laplacian spectral clustering, with the number of groups chosen by the eigengap, finds three communities: the doctors with the patients they visit, a large nurse group, and a small nurse and patient group. They mix roles, and the adjusted Rand index against roles is only about 0.2. Every top-ten pick of eigenvector and harmonic centrality sits in the large nurse group, while degree and betweenness reach two communities. That fits the idea that at ten picks those two rules isolate people who meet the same contacts, although we did not test that mechanism directly. Second, as an extension, we rewired the network twenty times, keeping everyone's number of contacts but scrambling who meets whom and reassigning the weights. That removes the community structure: the modularity of our three groups drops from 0.28 to about zero, although around 60 percent of edges stay in place. In those networks outbreaks are much larger, and degree is best or tied at every k. So the advantage of the weighted measures at small k depends on the ward's real contact pattern, not just on its degrees. Because rewiring also shuffles the weights, this does not prove that communities alone are responsible. Hasini will now discuss what remains open.

**Handover / cue line:** Hasini will now discuss what remains open.

### Slide 10: Open questions, and how to address them

**Speaker:** Hasini Kavuru · **94 words, about 0.7 min** (cue in notes: Hasini (about 40 seconds), then Arvinder (about 60 seconds))

Several questions remain open, and for each one we have a concrete way to address it. First, our rules pick the top k people one by one, which can mean several nurses who meet the same contacts. A greedy rule that adds, at each step, the person whose isolation most lowers the simulated attack rate would test whether picking sets does better. Second, we aggregated contacts into a static network and ignored their timing. A temporal SIR run on the time-stamped records would show whether that changes which rule is best. Arvinder will finish.

**Handover / cue line:** Arvinder will finish.

### Slide 10: Open questions, and how to address them

**Speaker:** Arvinder Singh Mundra · **144 words, about 1.0 min** (cue in notes: Hasini (about 40 seconds), then Arvinder (about 60 seconds))

Third, we hid contacts at random, but in real tracing people forget brief contacts first, so a natural next step is to hide the shortest contacts first and see whether the rules still hold up. And we should be clear about the limits: this is one ward over four days, the epidemic model is illustrative rather than calibrated, and isolated staff are not replaced. To wrap up, three takeaways. First, any centrality rule is much better than random isolation: with ten people isolated, there are 37 to 73 percent fewer infections at low and medium transmission. Second, with only a few isolation slots, the measures based on contact time do best; with ten slots, plain degree does as well or better. Third, missing contact data changes who gets chosen, but hardly how well isolation works. Thank you, and we are happy to take questions.

**Handover / cue line:** Thank you, and we are happy to take questions.

## Rehearsal checklist

- Time one full run-through with a stopwatch; aim for 12.5 minutes. If over 13, shorten the notes of slides 3 and 9 first.
- Each speaker says the handover line so the next person knows to start.
- Everyone should be able to answer any question in QA_Prep, not only their own.